

A Data-Driven Comparison of Car-Following Behavior Between Autonomous and Human-Driven Vehicles

Pedram Beigi

Department of Civil and Environmental Engineering,
The George Washington University,
800 22nd Street NW Washington, DC 20052
Corresponding Author: beigi@gwu.edu

Mohammad Emad Rashidi

Department of Civil and Environmental Engineering,
The George Washington University,
800 22nd Street NW Washington, DC 20052

Nachuan Li

Northwestern University Transportation Center,
600 Foster St., Evanston, IL 60208

Shayan Bafandkar

Department of Civil and Environmental Engineering,
University of Illinois at Urbana-Champaign,
205 North Mathews Ave., Urbana, IL 61801

Dana Monzer

Northwestern University Transportation Center,
600 Foster St., Evanston, IL 60208

John Hourdos

U.S. Department of Transportation, Federal Highway Administration
Turner-Fairbank Highway Research Center
6300 Georgetown Pike, McLean, VA 22101

Hani Mahmassani[†]

Northwestern University Transportation Center,
600 Foster St., Evanston, IL 60208

Alireza Talebpour

Department of Civil and Environmental Engineering,
University of Illinois at Urbana-Champaign,
205 North Mathews Ave., Urbana, IL 61801

Samer Hamdar

Department of Civil and Environmental Engineering,
The George Washington University,
800 22nd Street NW Washington, DC 20052

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[†]Hani Mahmassani is deceased.

ABSTRACT

This study examines longitudinal car-following behavior across small vehicles, large vehicles, and autonomous vehicles using high-resolution trajectory data and two representative models: the Intelligent Driver Model (IDM) and a Prospect Theory-based (PT) model. Model parameters are calibrated at the individual leader-follower level, and trajectory-level performance is evaluated using Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) metrics. To assess behavioral differences across vehicle types, statistical comparisons are conducted using Welch's ANOVA and Games-Howell post-hoc tests, which are robust to unequal variances and multiple pairwise comparisons. The results indicate that IDM provides strong predictive performance across all vehicle types, with high accuracy for autonomous vehicles. Error metrics show lower RMSE and MAE values for autonomous vehicles relative to human-driven vehicles. Statistical analysis reveals significant differences in IDM parameters related to safe time headway, desired speed, and minimum stopping distance, with large vehicles exhibiting longer following distances and lower desired speeds. These findings suggest that IDM parameters capture interpretable and physically meaningful distinctions across heterogeneous vehicle classes. In contrast, the PT model exhibits fewer statistically significant differences across vehicle types. While overall model fit remains high, Games-Howell tests identify a significant difference only in the maximum anticipation time horizon between small vehicles and autonomous vehicles. Other PT parameters show no significant variation across vehicle categories, indicating limited behavioral differentiation. This study highlights the strengths and limitations of deterministic and behaviorally motivated car-following models in mixed traffic. The resulting class-specific parameters offer actionable insights for practitioners developing lane-management and AV-integration strategies.

Keywords: Car-following, Prospect Theory, Autonomous Vehicles, Naturalistic Traffic, Trajectory Data

I. INTRODUCTION

Microscopic simulation serves as a pivotal tool in the analysis and design of traffic systems. It offers a platform for examining various traffic scenarios in a controlled environment, thereby avoiding disruptions to actual traffic flows [1]. It also allows studying the adoption of new technological systems whether in traffic control or communications. Car following (CF) models represent the core tasks in traffic flow, aiming to accurately replicate the longitudinal movement of vehicles [2] while following a leading vehicle and playing critical roles in the development of traffic flow theories, operations, and simulations. Over recent decades, numerous CF models have been developed to capture the dynamics of traffic behavior across a spectrum of conditions, highlighting their significance in traffic operations and safety. Despite significant advancements in modeling CF behaviors, our understanding of these crucial aspects remains incomplete, hindered by challenges in data collection, model calibration and validation, and the consideration of human factors [3] and mainly the limited availability of naturalistic large datasets.

Most tools for microscopic simulation leverage ordinary difference equations with a set of parameters to simulate drivers' behavior in car-following situations. Notably, in certain models, some parameters directly correspond to physical concepts in the real world [4]. Moreover, macroscopic traffic flow characteristics, such as free flow velocity, critical density, and jam density, are vital in numerous models. This is why the range of calibrated parameters are often significantly influenced by the characteristics of each traffic facility. Nonetheless, many car-following models include parameters that cannot be straightforwardly deduced from macroscopic observations [5], [6]. Consequently, there has been a growing interest in calibrating car-following models using empirical vehicle trajectory data which not only provides detailed insight into driving behaviors at the microscopic level [7] but also serves as the foundation for broader network trajectory propagation and data mining efforts [8]. Furthermore, driver behavior varies significantly across vehicle types (such as passenger cars and heavy vehicles) [9], roadway geometries, environmental conditions (like weather) [10], and the advent of Autonomous Vehicles (AVs). The introduction of AVs adds a new dimension to understanding vehicle behavior, distinguishing between autonomous and Human-Driven Vehicles (HDVs). Autonomous vehicles are expected and demonstrated to increase local traffic throughput and improve traffic stability. AVs have attracted significant attention in recent years because of their potential to revolutionize transportation safety and mobility [11]. However, while AV longitudinal control has been studied extensively, there remains limited empirical clarity on how AVs interact with surrounding heterogeneous traffic (e.g., heavy vehicles and human-driven passenger vehicles) and how these interactions translate into transferable, calibrated car-following parameters for mixed-flow simulation—particularly given platform-dependent “black-box” control logic. [12].

Hence, this study aims to leverage a naturalistic trajectory dataset [13] to calibrate the car-following models through the use of empirical trajectory data in an uncontrolled environment. The numerical findings allow analyzing and comparing the behaviors exhibited by a diverse array of vehicle categories. In particular, in this manuscript, we classify and designate the observed vehicles into i) small vehicles (e.g., passenger cars, Sports Utility Vehicles – SUVs and pick-up trucks), ii) large vehicles (e.g., trucks and buses among other heavy vehicles), and iii) autonomous vehicles, to gain a deeper understanding of their respective dynamics within complex freeway settings in suburban and urban environments.

This paper contributes to the field by: (1) Calibrating two distinct car-following models, the physics-based IDM and the behaviorally-driven Prospect Theory-based (PT) model, using a large-scale trajectory dataset (TGSIM); (2) Performing a comparative analysis of car-following behavior across three distinct vehicle classes: small vehicles, large vehicles, and AVs; (3) Utilizing statistical analysis to identify significant differences in driving parameters among these vehicle classes.

II. LITERATURE REVIEW

Numerous studies have been conducted to assess the behavior of vehicles following each other under various conditions. Among the physics-based and rule-based models, IDM [14] displayed more stable and uniform trajectories with consistent time headways and depicted realistic fundamental diagrams featuring stable and unstable regions [15]. Notably, IDM model captured traffic instability during congestion by allowing for more uniform and larger space headways. Also, demonstrated that among the sampled Shanghai drivers, the IDM exhibited the most favorable performance, with the lowest errors observed during both calibration and validation phases compared to other models [16]. In Pourabdollah et al. [17], three car following models commonly used in traffic simulation (i.e., the IDM, the Krauss car following model, and the Wiedemann car following model), are calibrated and verified on around 200 recorded trips. To find the optimal parameters of the car following models, a Genetic Algorithm (GA) minimizes a performance index which is derived to reflect the error in the vehicle's impact on traffic flow as well as its energy demand, compared to a reference trip. The result of the simulation shows that on average, the optimized IDM can replicate driving behaviors best compared to the other two models. In Chen et al. [18], the authors studied the car-following behaviors of individual drivers in real traffic scenes using the trajectory datasets provided by Next Generation SIMulation (NGSIM) program. This study calibrated IDM and Microscopic Traffic SIMulator (MITSIM) car-following models by using GA.

Additionally, lots of studies have tried to incorporate behavioral components. The psychological-based models that incorporate parameters that reflect human driver characteristics and interactions, are suitable for simulating traffic scenarios involving human drivers. The Prospect Theory (PT) model exhibits behavioral realism, capturing the nuanced decision-making processes of drivers. Particularly effective in uncertain conditions, the PT model accounts for the inherent unpredictability of real-world driving scenarios. Adewale et al. [19] extends a data-driven car-following model to incorporate drivers' memory, and cooperation with the lead vehicle. The model predicts the following vehicle's speed in AV-HDV mixed traffic. The effect of drivers' cooperation on car-following behavior was modeled using PT, whereas the driver's memory was incorporated using the memory cell of a long short-term memory (LSTM) neural network. This extended car-following model is called the "PT-LSTM model." Real-world vehicle trajectories of HDVs and AVs in the Waymo AV Open Dataset were used to calibrate and validate the PT-LSTM model. The PT-LSTM model demonstrated higher accuracy compared with the LSTM model. Ding et al. [20] also developed an extended IDM for AVs and HDVs that incorporated cooperation between drivers, modeled using PT. Unlike other studies that made use of simulated data, they used the field data observed from three Tesla vehicles in mixed traffic flow. The results indicated that the model captured the heterogeneity of car-following behaviors in AV-HDV mixed traffic. Sharma et al. [21] incorporates the driver compliance behavior into a connected vehicle driving strategy (CVDS) that can be integrated with traditional car-following (CF) models to better

describe the connected vehicle CF behavior. Driver compliance, a key human factor for the success of connected vehicles technology, is modelled using a celebrated theory of decision making under risk – the PT. The reformulated value and weighting functions of PT are consistent with the driver compliance behavior and also preserve the integral elements of PT. Furthermore, the connected vehicle trajectory data collected from a carefully designed advanced driving simulator experiment are utilized to calibrate CVDS integrated with IDM, i.e., CVDS-IDM. The calibration results reveal that drivers in the connected environment drive safely and efficiently. Moreover, the CVDS-IDM can successfully model and predict the CF dynamics of connected vehicles and is more behaviorally and numerically sound than a traditional CF model. Sharma et al. developed an extended IDM for connected vehicles that incorporated driver compliance with information [19]. The driving strategy developed using PT was used for calibrating this. Sharma et al. found that the extended IDM better predicted car-following behavior than the original. Hamdar et al. [22] explore a PT-based approach to model driver car-following behavior, focusing on capturing fundamental diagram regions and safety indices. The model combines subjective utilities for accelerations and disutility for decelerations with the perceived risk of rear-end collisions, grounded in prospect theory. Calibration and validation are performed using microscopic trajectory data with a modified genetic algorithm to account for inter-driver heterogeneity. Simulation sensitivity analysis reveals the model's ability to capture traffic breakdowns, instabilities in congested regions, and the impact of psychological factors and perception errors on accident distribution.

Several papers have investigated the behavior of Adaptive Cruise Control (ACC) vehicles which have the highest deployment among the various AV levels. Most of these studies have looked into experimental field data. In [23] the study looked into a platoon of a human vehicle lead with specific speed profiles, followed by two ACC vehicles. The study used the Asymmetric Behavior model which is an extension of Newell's model [24] to analyze ACC's response to different headway settings, traffic speed, and stimulus from the leading vehicle. The study showed that ACC response was comparable to human vehicles but larger than AV controller time gap and that the headway has a negative effect on ACC deceleration rate change. In [25] experimental field data was collected using a platoon of EV ACC vehicles to verify the argument that the latter have benefit over internal combustion engine ACC vehicles in terms of reducing queue discharge delays. The data by that study has been analyzed by other papers, such as [26], which developed the EVM car-following model for EV ACC vehicles, and showed that EVM does a better job in modeling the initial acceleration after braking as compared to IDM and OVRV despite all three models being sufficiently accurate enough in modeling other aspects of the EV ACC vehicle's behavior. On the contrary, several papers have shown that some versions of IDM do a better job in modeling ACC vehicles than other models. In [27], it was shown that Improved IDM with Constant-Acceleration Heuristic works best for fitting the ideal AV trajectories based on generated trajectories. In [28] it is shown that enhanced IDM for ACC vehicles outperform PATH-ACC and Continuous Asymmetric Optimal Velocity Relative Velocity. The paper uses the OpenACC dataset for validation [29] where a platoon of ACC vehicles follows a human vehicle in a controlled experiment.

The calibration of CF behaviors based on trajectory data for AVs in a mixed environment with HDVs remains an underexplored area. The work presented in [30] provides calibration for both the car-following and lane-changing models using identical field data, demonstrating that driver decisions may not always be rational in certain scenarios. In [2], the focus is on calibrating two advanced car-following models with empirical trajectories that reflect the multi-behavioral

patterns of ACC-equipped vehicles. The findings contribute to a more profound understanding of commercial ACC systems and their emulation within microsimulation frameworks. Significantly, the calibration of multi-behavioral patterns not only enriches the calibration's physical significance for ACC-equipped vehicles but also facilitates a balanced performance across different behavioral patterns, thereby enhancing the comprehension of their benefits and limitations in capturing specific behaviors. Hu et al. [31] seeks to delineate the empirical car-following behaviors of Waymo's AV and to contrast these with those of HDVs, utilizing the IDM for this purpose. The key insights include: (a) AVs demonstrate a safer operational mode than HVs, as evidenced by longer time headways and greater jam spacings; (b) AVs exhibit a markedly longer response time to diverse stimuli compared to HVs; (c) despite the limited trajectory data from the Waymo Open Dataset, its suitability for calibrating certain IDM parameters is verified, with calibration outcomes aligning with empirical analyses. Furthermore, the analysis suggests a lower propensity for string unstable behavior in AVs compared to HVs, and (d) the study observes minimal differences between HVs following AVs versus following other HVs, aside from reduced jam spacing in the former scenario. The overarching conclusion is that AVs currently adopt a conservative approach prioritizing safety, albeit at the expense of traffic flow efficiency. The research in [12] involves calibrating the IDM for mixed autonomy traffic using real-world trajectory data. This study introduces an adaptive variant of the IDM that allows for real-time adjustments of its parameters according to the current traffic conditions. Additionally, incorporating the standard deviation of velocity into the calibration objective function aids in capturing stop-and-go traffic behaviors. The outcomes indicate that the adaptive IDM, along with the calibration methodology, effectively replicates observed traffic patterns in mixed autonomy environments.

III. METHODOLOGY

DATA

This study is based on naturalistic data, TGSIM data [13]. This data focus on data collection efforts for automated vehicles across various locations, focusing on different levels of automation and traffic conditions. In Chicago, IL, two datasets were collected on I-90/I-94, involving major weaving sections and segments with diverse ramp configurations, covering areas between N Kimball Ave. and N Wilson Ave., and featuring 2.5 and 0.8 miles of highway driving. One data in near Hinsdale, IL, on I-294, has been collected focusing on automated vehicles with an emphasis on heavy vehicle traffic over 3 miles of highway, showcasing a major on-ramp and several off-ramps. In Washington D.C., on I-395, the effort captured a 0.34-mile segment of urban expressway with significant weaving and lane-changing challenges [32]. These efforts provide comprehensive insights into automated vehicle behavior in diverse infrastructural and traffic conditions.

For the calibration of the models, we first process the data according to the following conditions, the duration of the car-following must be greater than 10 seconds, and the distance headway should be less than 200 meters; and also there should be no lane-changing behavior during the car-following process, and the ID of the preceding vehicle monitored during this time must remain unchanged.

1 IDM MODEL

2 The IDM model [33], [34] that is widely used in transportation research to model longitudinal
 3 movement has been selected to describe the car-following characteristics of human-driven
 4 vehicles. . The model is based on a set of equations that describe the acceleration of a vehicle as a
 5 function of its current speed, the desired speed, headway, and the relative speed with the leading
 6 vehicle. The IDM acceleration equation is given by:

$$7 \quad \dot{v}_n(t) = a \left(1 - \left(\frac{v_n(t)}{v_0} \right)^\delta - \left(\frac{s(v_n(t), \Delta v_n(t))}{s_0} \right)^2 \right) \quad (1)$$

8 Where $\dot{v}_n(t)$ is the acceleration of the vehicle at time t , a is a constant representing the
 9 maximum acceleration, $v(t)$ is the speed of the vehicle at time t , v_0 is the desired speed, δ is a
 10 calibration parameter, and $s(v(t), \Delta v(t))$ is the desired gap to the leading vehicle, incorporating
 11 the relative speed $\Delta v(t)$ and additional parameters. The desired gap $s(v(t), \Delta v(t))$ is calculated
 12 using the equation:

$$13 \quad s(v_n(t), \Delta v_n(t)) = s_0 + v_n(t) \cdot T + \frac{v_n(t) \Delta v_n(t)}{2\sqrt{ab}} \quad (2)$$

14 Where s_0 is the minimum bumper-to-bumper distance, T is the safe time headway, $\Delta v(t)$
 15 is the relative speed to the leading vehicle, and a, b are parameters influencing the acceleration
 16 term.

18 PROSPECT THEORY MODEL

19 The Prospect Theory-based car-following model [35] integrates the principles of Prospect Theory
 20 into traditional car-following models to better capture driver behavior under uncertainty. This
 21 model accounts for drivers' psychological biases, such as loss aversion and reference dependence,
 22 in their response to the vehicle ahead. Unlike conventional models that assume rational and
 23 consistent driver responses, the PT-based model posits that drivers perceive potential gains and
 24 losses relative to a reference point, often their current state or expectations. Drivers exhibit loss
 25 aversion, reacting more strongly to perceived losses (e.g., the risk of a collision or increased travel
 26 time) than to equivalent gains (e.g., smoother flow or reduced travel time). This approach allows
 27 for a more realistic representation of driver decision-making processes, especially in complex and
 28 uncertain traffic conditions, by considering how drivers' risk preferences and subjective
 29 perceptions influence their acceleration and deceleration behaviors. The (generalized) utility U of
 30 the model is composed of the generalized (or perceived) prospect-theoretic acceleration utility
 31 $U_{PT}(a)$ whose form is derived by the prospect theory, and a penalty $U_{crash}(a)$ for the risk of
 32 accidents:

$$33 \quad U(a; s; v; \Delta v) = U_{PT}(a) + U_{crash}(a; s; v; \Delta v) \quad (3)$$

34 The utility is specified below where $x = \frac{a}{a_0}$

$$35 \quad U_{PT}(x) = x[0.5w_m + 0.5(1 - w_m)(\tanh(x) + 1)](1 + x^2)^{0.5(\gamma-1)} \quad (4)$$

$$36 \quad U_{crash}(a; s; v; \Delta v) = P_c W_c \quad (5)$$

where γ (Gamma) is sensitivity exponents of the generalized utility, w_m asymmetry factor for negative utilities, α is speed uncertainty variation coefficient, W_c weighing factor for accidents, τ maximum anticipation time horizon, β logit uncertainty parameter (intra-driver variability), τ_{corr} correlation time of intra-driver variability, a_{max} maximum acceleration, v_0 desired speed, and s_0 minimum gap. The final acceleration is determined from a probability density function, reflecting the stochastic nature of a driver's response to stimuli. Incorporating the [36] formulation, the methodology begins by defining the effective gap S_{eff} as the difference between the actual gap and a defined minimum gap S_o . This is expressed as:

$$S_{eff} = (gap - S_o, 0.1) \quad (6)$$

The time gap τ is calculated based on the vehicle's relative speed Δv and the effective gap S_{eff} . If the relative speed is greater than the ratio of the effective gap to the maximum time gap T_{max} , then τ is given by:

$$\tau = \frac{S_{eff}}{\Delta v} \quad (7)$$

Otherwise, it is set to T_{max}

$$\tau = T_{max} \quad (8)$$

To avoid division by zero, a small value is added if $\Delta v = 0$ and if $\alpha = 0$, ensuring the stability of the calculations. Then, if the $\ln \ln (W_c \cdot Z')$ is greater than or equal to zero, the intermediate calculation parameters Z' and Z^* are then computed as:

$$Z' = \frac{\tau}{2\alpha \cdot \Delta v} \quad (9)$$

$$Z^* = -\frac{\sqrt{2 \ln \ln (W_c \cdot Z')}}{\sqrt{2} \pi} \quad (10)$$

Otherwise, Z^* is set to zero. The initial acceleration value A^* is then determined using the formula:

$$A^* = \frac{2}{\tau} \left(\frac{S_{eff}}{\tau} - \Delta v + \alpha \cdot \Delta v \cdot Z^* \right) \quad (11)$$

An iterative Newton-Raphson method is applied to refine A^* . During each iteration, the derivative functions U' and U'' are updated depending on the sign of A^* . If $A^* \geq 0$:

$$U' = \gamma_1 \cdot (A^*)^{\gamma_1-1} \quad (12)$$

$$U'' = \gamma_1 \cdot (\gamma_1 - 1) \cdot (A^*)^{\gamma_1-2} \quad (13)$$

Otherwise:

$$U' = W_m \cdot \gamma_2 \cdot (-A^*)^{\gamma_2-1} \quad (14)$$

$$U'' = -W_m \cdot \gamma_2 \cdot (\gamma_2 - 1) \cdot (-A^*)^{\gamma_2-2} \quad (15)$$

The parameter Z is recalculated within each iteration as:

$$Z = \frac{\Delta v + 0.5 \cdot A^* \cdot \tau - \frac{S_{eff}}{\tau}}{\alpha \cdot \Delta v} \quad (16)$$

and the cumulative distribution function value F is computed using the normal distribution. The function F and its derivative F' are given by:

$$F = U' - W_c \cdot F(Z) \cdot Z' \quad (17)$$

$$F' = U'' - W_c \cdot F(Z) \cdot (Z \cdot (Z')^2 + Z'') \quad (18)$$

The value of A^* is updated iteratively:

$$A^* = A^* - \frac{F}{F'} \quad (19)$$

Finally, the conditional car-following acceleration a_{cf} and the free-flow acceleration a_{ff} are calculated. The variance σ^2 and a random Wiener process value Y_t are used to determine a_{cf} :

$$a_{cf} = A^* + \sigma^2 \cdot Y_t \quad (20)$$

where Y_t is given by:

$$Y_t = \exp \exp \left(-\frac{0.1}{\tau} \right) + \sqrt{\frac{24 \cdot 0.1}{\tau}} \cdot Wei_t \quad (21)$$

Here, Wei_t would be a random Wiener sample from a standard normal distribution to incorporate random fluctuations in the driver's acceleration behavior. The free-flow acceleration is given by:

$$a_{ff} = a_{max} \left(1 - \frac{speed}{v_{desired}} \right) \quad (22)$$

The final acceleration a is the minimum of a_{cf} and a_{ff} , bounded by a maximum of 3 m/s² and a minimum of -5 m/s²:

$$a = (a_{cf}, a_{ff}) \quad (23)$$

$$a = ((a, 3), -5) \quad (24)$$

GENETIC ALGORITHM

Calibration of traffic models is essential to ensure accurate representation of real-world traffic dynamics [1]. This process involves finely adjusting model parameters to closely match observed data collected from the field. Genetic Algorithms (GA) are effective for this purpose, leveraging principles from natural selection and genetics to optimize complex models efficiently across large parameter spaces, avoiding local optima.

The Genetic Algorithm (GA) utilized for calibrating is a bio-inspired optimization technique that mimics the process of natural evolution. This algorithm is particularly suited for solving complex optimization problems where the search space is vast and not well understood a priori. In the context of calibration, the objective is to find a set of parameters $\theta_{i,IDM} = \{T, a, b, v_0, \delta, s_0\}$ for IDM, and $\theta_{i,PT} = \{T_{max}, \alpha, \beta, W_c, W_m, \gamma\}$ for PT, that minimizes the

discrepancy between observed vehicle behaviors and those predicted by the models. The GA method begins with an initial population of candidate solutions, where each individual represents a potential set of parameters θ_i . The fitness of each individual is evaluated through a fitness function, defined as:

$$f(\theta_i) = \frac{1}{1 + \sum_j (\lambda_p |p_{obs,j} - p_{sim,j}(\theta_i)| + \lambda_v |v_{obs,j} - v_{sim,j}(\theta_i)|)} \quad (25)$$

where λ_p and λ_v are scaling factors to balance the contribution of the position and velocity errors in the fitness function; $p_{obs,j}$ and $v_{obs,j}$ are the observed positions and velocities at time j , and $p_{sim,j}(\theta_i)$ and $v_{sim,j}(\theta_i)$ are the simulated positions and velocities using the parameters θ_i . This fitness function aims to minimize the sum of absolute differences between observed and simulated positions and velocities, thus guiding the selection of parameters that best replicate real-world driving behavior.

TABLE 1: List of Calibration Parameters

Model	Model parameter	Typical Value [4][22]	Range of variation
IDM	T safe time headway (s)	0.8 to 2	(0.5, 2.5)
	a starting acceleration (m/s^2)	0.8 to 2.5	(0.3, 5.0)
	b comfortable deceleration value (m/s^2)	2	(0.5, 3.0)
	v_0 free slow speed (m/s)	---	(5, 35)
	s_0 safe stopping distance bumper-to-bumper (m)	2	(1, 5)
	δ Acceleration exponent	4	(3.8, 4.2)
PT	τ maximum anticipation time horizon (s)	5.0	(2, 8.0)
	α speed uncertainty variation coefficient (-)	0.08	(0, 0.6)
	β logit uncertainty parameter (intra-driver variability) (-)	5.4	(2, 8)
	w_c weighing factor for accidents (-)	97500	(6E4, 1.3E5)
	W_m asymmetry factor for negative utilities (-)	3.85	(2, 8.0)
	γ sensitivity exponents of the generalized utility (-)	0.32	(0.3, 2.0)

IV. MODEL CALIBRATION AND BEHAVIORAL ANALYSIS

A. Car-Following Model Calibration

As noticed by most researchers, since different drivers certainly have diverse driving habits, and their driving situation may also be different, the obtained model parameters vary from one dataset to another [37]. A set of overall parameters may lead to giant error when applied in some specific drivers. Thus, we choose to calibrate the parameters with each leader-follower pair respectively. Also, the number of AVs and other small and large vehicles in the dataset are not the same. Therefore, we have selected an equal number of small and large vehicles based on the number of AVs available in the dataset. To ensure that the chosen small and large vehicles represent the same environment as the AVs, we selected surrounding vehicles based on the location and time of the

AVs' trajectories. Consequently, with the presence of 120 AVs in the data, 120 small and 120 large vehicles were also chosen for calibration purposes. The ranges chosen for each parameter are based on the literature [22], [38]. In addition after few testing $\lambda_p = 1$ and $\lambda_v = 0.5$ were chosen. FIGURE 1 displays the simulated versus observed locations and speeds of a sample follower. Table 2 summarizes the episode-level calibration results by vehicle class, reporting the mean $\pm$ standard deviation (median in parentheses) of the calibrated parameters across the 120 leader-follower episodes in each class.

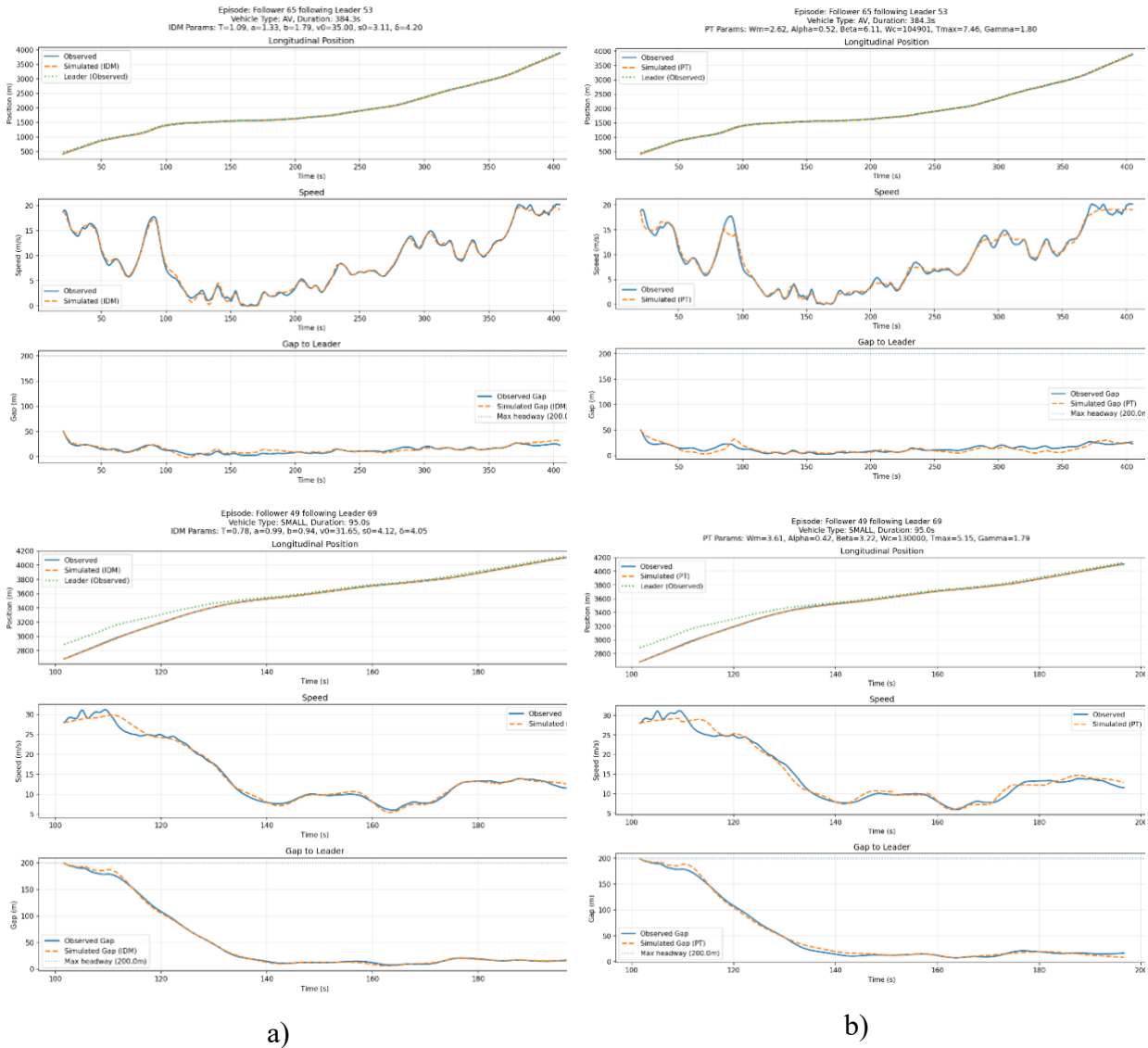


FIGURE 1: Simulated (non-smooth) versus observed position and speeds for two sample leader and follower for a) IDM and b) PT

TABLE 2: Car Following parameters calibration results

	Parameter	Small Vehicles Mean $\pm$ SD (Median)	Large Vehicles Mean $\pm$ SD (Median)	Autonomous Vehicles Mean $\pm$ SD (Median)
IDM	T	1.073 ± 0.705 (0.786)	1.462 ± 0.836 (1.298)	1.059 ± 0.667 (0.785)
	a	1.589 ± 1.552 (0.935)	1.572 ± 1.687 (0.737)	1.261 ± 1.319 (0.692)
	b	1.631 ± 1.145 (1.166)	1.644 ± 1.157 (1.140)	1.906 ± 1.088 (2.245)
	v_0	28.085 ± 9.451 (34.860)	25.015 ± 9.743 (26.632)	29.546 ± 7.817 (34.959)
	s_0	2.879 ± 1.803 (2.309)	3.414 ± 1.656 (3.894)	3.425 ± 1.691 (4.243)
	δ	4.064 ± 0.173 (4.199)	4.027 ± 0.180 (4.110)	4.052 ± 0.162 (4.124)
PT	T_{max}	6.304 ± 1.684 (6.785)	6.481 ± 1.759 (7.235)	6.797 ± 1.539 (7.517)
	α	0.452 ± 0.172 (0.533)	0.423 ± 0.185 (0.485)	0.460 ± 0.179 (0.556)
	β	5.003 ± 1.860 (5.003)	4.829 ± 1.887 (4.895)	4.824 ± 1.914 (4.851)
	W_c	95095.173 ± 24011.861 (93417.592)	96985.148 ± 22439.055 (98096.735)	99323.384 ± 23180.168 (102710.675)
	W_m	5.148 ± 1.962 (5.280)	5.188 ± 1.951 (5.171)	5.398 ± 1.906 (5.597)
	Γ	1.338 ± 0.494 (1.704)	1.347 ± 0.497 (1.679)	1.456 ± 0.463 (1.731)

B. Comparative Analysis of Vehicle Behaviors

When observing the calibration results in Table 2, distinct driving styles emerge across vehicle types. For the IDM, large vehicles exhibit the highest safe time headway ($T = 1.462$ s), indicating they maintain larger gaps, which is consistent with their longer braking distances. Autonomous vehicles show a slightly lower time headway ($T = 1.059$ s) than small vehicles ($T = 1.073$ s), suggesting that their following behavior is comparable and not substantially more conservative in terms of spacing. Conversely, while autonomous vehicles appear to have a slightly higher mean comfortable deceleration ($b = 1.906$ m/s²) and small vehicles a higher mean starting acceleration ($a = 1.589$ m/s²), subsequent statistical testing reveals these parameters are highly variable and do not statistically differ across the fleet. For the PT model, autonomous vehicles exhibit the highest maximum anticipation time horizon ($T_{max} = 6.797$ s). While their mean sensitivity exponent ($\Gamma = 1.456$) also appears marginally higher, intra-driver variability dominates this parameter, making the differences statistically insignificant.

To compare the performance of calibrated models for different types of followers, their respective errors are used. By evaluating the discrepancies between the predicted and observed vehicle trajectories, the comparison focuses on key metrics such as root mean square error (RMSE), and mean absolute error (MAE) to determine which model provides a more accurate representation of real-world driving behavior. Table 3 presents the mean values of RMSE, and MAE for different vehicle types.

An analysis of the error metrics in Table 3 provides insight into how each model's performance varies by vehicle type. For the IDM model, the calibration achieves the best fit for autonomous vehicles. This is evidenced by the IDM having the lowest Root Mean Square Error (RMSE) of 2.002 and the lowest Mean Absolute Error (MAE) of 1.491 for autonomous vehicles

compared to other vehicle types. This suggests that the IDM's deterministic, rule-based framework aligns effectively with the control logic observed in the autonomous vehicles from the dataset.

Also, the PT model's calibration results are most accurate for autonomous vehicles, as indicated by the lowest RMSE (2.919) and lowest MAE (2.251) among the vehicle categories. This suggests that the PT model is capable of capturing the longitudinal dynamics of autonomous vehicles with high fidelity, although its performance for human-driven vehicles remains comparable across small and large vehicle classes.

TABLE 3: Summary of error metrics

Model	Vehicle Type	RMSE	MAE
IDM	Small Vehicles	2.979	2.266
	Large Vehicles	3.870	3.001
	Autonomous Vehicles	2.002	1.491
PT	Small Vehicles	3.357	2.588
	Large Vehicles	3.347	2.612
	Autonomous Vehicles	2.919	2.251

C. Statistical Validation of Behavioral Differences

To determine whether there are statistically significant differences in the parameters across different types of vehicles, which can provide insights for optimizing vehicle behavior models and improving traffic management strategies, Welch's ANOVA is used to compare the means of three or more groups under potentially unequal variances [39], and the Games-Howell post-hoc test is used to perform pairwise comparisons between vehicle types without assuming equal variances or equal sample sizes, ensuring a robust statistical analysis of vehicle parameters across categories [40]. Since Welch's ANOVA and Games-Howell procedures are designed to be robust under heteroscedasticity, equal-variance assumptions were not required.

Welch's ANOVA is a statistical method used to determine whether there are significant differences between the means of three or more independent groups when the assumption of equal variances may not hold. Similar to classical ANOVA, the method compares the variability between groups to the variability within groups; however, it accounts for heterogeneity in group variances by weighting each group according to its sample variance. If the adjusted between-group variability is significantly larger than the within-group variability, it suggests that at least one group mean differs from the others. The Welch F-statistic is computed as follows:

$$F_W = \frac{\sum_{i=1}^k \omega_i (\bar{x}_i - \bar{x}_\omega)^2 / (k - 1)}{1 + \frac{2(k - 2)}{k^2 - 1} \sum_{i=1}^k \frac{\left(1 - \frac{\omega_i}{W}\right)^2}{n_i - 1}}$$

Where:

$\omega_i = \frac{n_i}{s_i^2}$ is the weight of group i ,

n_i and s_i^2 are the sample size and variance of group i ,

1 x_i is the mean of group i ,

2 $\bar{x}_\omega = \frac{\sum_{i=1}^k \omega_i \bar{x}_i}{\sum_{i=1}^k \omega_i}$ is the weighted mean,

3 $W = \sum_{i=1}^k \omega_i$ and,

4 k is the number of groups.

5 The degrees of freedom associated with the Welch F-statistic are approximated using the
 6 Welch–Satterthwaite equation. A significant Welch F-statistic indicates that there are statistically
 7 significant differences among the group means. Table 4 presents the results of the Welch ANOVA
 8 performed on the vehicle parameters (T , a , b , v_0 , s_0) across three vehicle types: small vehicles,
 9 large vehicles, and autonomous vehicles. The reported F-values and p-values indicate whether
 10 significant differences exist for each parameter, with statistical significance assessed at the 0.05
 11 level.

12 The Games–Howell post-hoc test is used to determine whether there is a significant
 13 difference between the means of pairs of independent groups, particularly when the variances and
 14 sample sizes of the two groups are unequal and multiple pairwise comparisons are performed. The
 15 test statistic for the Games–Howell test is calculated as:

$$q = \frac{|X_1 - X_2|}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \quad (29)$$

17 The computed q value is evaluated against the studentized range distribution to assess
 18 statistical significance. Where X_1 and X_2 are the sample means, s_1^2 and s_2^2 are the sample
 19 variances, and n_1 and n_2 are the sample sizes of the two groups. The degrees of freedom for the
 20 test are calculated using the following formula, which accounts for unequal variances:

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}} \quad (30)$$

22 This test compares the means of the two groups and determines whether the difference is
 23 statistically significant. Unlike the traditional t-test, statistical significance is assessed using the
 24 Studentized range distribution, which allows the Games–Howell test to control for multiple
 25 comparisons without assuming equal variances or equal sample sizes. Table 5 summarizes the
 26 results of the Games–Howell post-hoc tests performed for pairwise comparisons of vehicle
 27 parameters (T , a , b , v_0 , s_0) between small vehicles (S), large vehicles (L), and autonomous vehicles
 28 (A). The test statistic and p-value are shown for each comparison, and the “Significance” column
 29 indicates whether the difference is statistically significant at the 0.05 level.

TABLE 4: Welch ANOVA Results for IDM Parameters Across Different Vehicle Types

Parameter	F-value	p-value	Significance
T	10.0885	6.25e-05	Significant Difference
a	2.0167	0.1354	No Significant Difference
b	2.3478	0.0978	No Significant Difference
v_0	7.9193	0.00047	Significant Difference
s_0	3.7568	0.0248	Significant Difference

TABLE 5: Games–Howell Post-hoc Results for Pairwise Comparisons of IDM Parameters

Parameter	Types	t-Statistic	df	p-Value	Hedges g	Significance
T	S vs L	3.8976	231.4	0.000373	0.50159	Significant Difference
	S vs A	-0.1602	237.3	0.985946	-0.0206	No Significant Difference
	L vs A	-4.1298	226.8	0.000151	-0.5315	Significant Difference
a	S vs L	-0.0794	236.4	0.996532	-0.0102	No Significant Difference
	S vs A	-1.7624	231.9	0.184614	-0.2268	No Significant Difference
	L vs A	-1.5911	224.9	0.251586	-0.2048	No Significant Difference
b	S vs L	0.08652	238.0	0.995881	0.01113	No Significant Difference
	S vs A	1.90934	237.4	0.138252	0.24572	No Significant Difference
	L vs A	1.80983	237.1	0.16855	0.23291	No Significant Difference
v_0	S vs L	-2.4780	237.8	0.036904	-0.3189	Significant Difference
	S vs A	1.30455	229.9	0.394081	0.16789	No Significant Difference
	L vs A	3.97366	227.3	0.000279	0.51138	Significant Difference
s_0	S vs L	2.3922	236.3	0.045970	0.30786	Significant Difference
	S vs A	2.42037	237.0	0.042801	0.31148	Significant Difference
	L vs A	0.05328	237.9	0.998436	0.00686	No Significant Difference

The analysis of vehicle parameters across different vehicle types using Welch ANOVA and Games–Howell post-hoc tests reveals several key findings. The Welch ANOVA and Games–Howell results in Tables 4 and 5 indicate statistically significant differences in several IDM parameters among vehicle types. The Welch ANOVA reveals significant differences in the mean safe time headway (T), desired speed (v_0), and minimum stopping distance (s_0) across the three groups, while no significant differences are observed for starting acceleration (a) and comfortable deceleration (b).

The Games–Howell post-hoc tests further clarify these differences. A significant difference in time headway exists between small and large vehicles ($p = 0.00037$), confirming that large vehicles maintain longer following distances, as well as between large and autonomous vehicles ($p = 0.00015$). For desired speed (v_0), significant differences are observed between small and large vehicles ($p = 0.0369$) and between large and autonomous vehicles ($p = 0.00028$), indicating that large vehicles operate at systematically lower desired speeds. In addition, the minimum stopping distance (s_0) shows significant differences between small vehicles and both large vehicles ($p =$

0.0460) and autonomous vehicles ($p = 0.0428$). The significantly larger minimum stopping distance for autonomous vehicles compared to small human-driven vehicles physically aligns with the conservative, safety-first programming typical of current commercial AV stacks, which enforce larger bumper-to-bumper standstill gaps to prevent low-speed rear-end collisions. Meanwhile, no significant pairwise differences are found for starting acceleration (a) or comfortable deceleration (b), suggesting that these parameters are more uniform across the observed vehicle fleet in this dataset.

TABLE 6: Welch ANOVA Results for PT Parameters Across Different Vehicle Types

Parameter	F-value	p-value	Significance
T_{max}	2.90728	0.0565732	No Significant Difference
α	1.38446	0.252471	No Significant Difference
β	0.352531	0.703273	No Significant Difference
W_c	0.967125	0.381664	No Significant Difference
W_m	0.584528	0.558167	No Significant Difference
γ	2.29267	0.103225	No Significant Difference

TABLE 7: Games–Howell Post-hoc Results for Pairwise Comparisons of PT Parameters

Parameter	Types	t-Statistic	df	p-Value	Hedges g	Significance
T_{max}	S vs L	0.79611	237.5	0.70578	0.1024	No Significant Difference
	S vs A	2.36666	236.1	0.04901	0.3046	Significant Difference
	L vs A	1.48074	233.9	0.30202	0.1906	No Significant Difference
α	S vs L	-1.2757	236.9	0.41033	-0.1642	No Significant Difference
	S vs A	0.34	237.7	0.93829	0.0438	No Significant Difference
	L vs A	1.58252	237.7	0.25518	0.2036	No Significant Difference
β	S vs L	-0.7170	238.0	0.75366	-0.0923	No Significant Difference
	S vs A	-0.7338	237.8	0.74365	-0.0944	No Significant Difference
	L vs A	-0.0219	237.0	0.99974	-0.0028	No Significant Difference
W_c	S vs L	0.62997	236.9	0.80379	0.0811	No Significant Difference
	S vs A	1.3878	237.7	0.34890	0.17860	No Significant Difference
	L vs A	0.793942	237.8	0.707117	0.10217	No Significant Difference
W_m	S vs L	0.158817	238.0	0.986192	0.02044	No Significant Difference
	S vs A	1.00294	237.8	0.575728	0.12907	No Significant Difference
	L vs A	0.84476	237.9	0.675571	0.10871	No Significant Difference
γ	S vs L	0.142261	238.0	0.988905	0.01831	No Significant Difference
	S vs A	1.90813	237.0	0.138598	0.24556	No Significant Difference
	L vs A	1.75712	236.8	0.187147	0.22587	No Significant Difference

For the PT model, the statistical tests in Tables 6 and 7 reveal more limited behavioral distinctions across vehicle types. The Welch ANOVA results indicate no statistically significant differences across the three vehicle categories for most PT parameters. The Games–Howell post-

hoc tests further clarify these findings. A significant difference is observed only for the maximum anticipation time horizon (T_{max}) between small vehicles and autonomous vehicles ($p = 0.0490$), with autonomous vehicles exhibiting a longer anticipation horizon. This suggests that AVs are modeled as slightly more forward-looking than small vehicles in terms of anticipatory behavior.

In contrast, no statistically significant pairwise differences are found for the logit uncertainty parameter (β), the sensitivity exponent (Gamma), or the remaining PT parameters, indicating that intra-driver variability and sensitivity to perceived gains and losses are broadly similar across vehicle types in this dataset. Overall, these results suggest that, unlike the IDM parameters, the PT model captures fewer pronounced behavioral differences between human-driven and autonomous vehicles under the current calibration framework.

V. CONCLUSION

This study investigated longitudinal car-following behavior across small vehicles, large vehicles, and autonomous vehicles by calibrating the Intelligent Driver Model (IDM) and the Prospect Theory-based (PT) model using high-resolution, naturalistic trajectory data. Model parameters were estimated at the individual leader–follower level, with model performance assessed through trajectory-based error metrics, specifically RMSE and MAE. Statistical comparisons were conducted using Welch ANOVA and Games-Howell post-hoc tests to identify significant behavioral distinctions across vehicle classes.

The calibration results demonstrate that the IDM provides a robust fit across all vehicle types, achieving the lowest RMSE and MAE for autonomous vehicles. This suggests that the IDM's deterministic, rule-based formulation aligns exceptionally well with the control logic observed in the AV trajectories. Furthermore, the statistical analysis of the IDM parameters captured interpretable, physically meaningful distinctions. For example, large vehicles maintained significantly longer time headways and lower desired speeds than other classes, reflecting their physical constraints. Autonomous vehicles exhibited significantly larger minimum stopping distances compared to small human-driven vehicles, aligning with the conservative, safety-first programming typical of current commercial AV stacks. However, it should be noted that the calibrated desired speed (v_0), while significantly different across vehicle classes, may act partially as a compensatory variable during trajectory-level optimization rather than an exact reflection of true free-flow intentions. In contrast, the PT model exhibited fewer statistically significant differences across vehicle types under the current deterministic calibration framework, pointing to a potential trade-off between behavioral richness and parameter identifiability.

The empirically calibrated longitudinal-control surrogates developed in this study offer practical applications for practitioners and policymakers. The statistically significant differences in headway and spacing parameters support targeted lane-management and speed-harmonization strategies in corridors with high heavy-vehicle shares. Furthermore, utilizing class-specific parameters helps agencies avoid overestimating or underestimating capacity and stability impacts during staged AV integration, supporting better scenario testing for safety margins and operational standards.

Despite these insights, this study has several limitations. First, the dataset is constrained to 120 samples per vehicle type. While this provides a sufficient baseline for trajectory-level optimization, larger sample sizes would enhance the generalizability of the findings. Second,

models were calibrated to the full episode to capture complete behavioral regimes, meaning rigorous cross-validation on split trajectories was not performed. Third, the route flow assignments in the simulation environment may not perfectly reflect real-world macroscopic conditions, necessitating caution when interpreting macro-level performance results. Finally, because AV longitudinal control logic varies substantially across manufacturers and software versions, direct transfer of the estimated parameters to different regions should be performed carefully; these estimates are best utilized as informed priors for local calibration.

Future research should extend this work by stratifying car-following dynamics by congestion regime (e.g., free-flow versus stop-and-go) to capture regime-dependent parameter shifts. Additional extensions leveraging this naturalistic dataset include calibrating lane-changing behaviors by vehicle class, quantifying multi-vehicle interactions such as short-horizon platooning, and incorporating stochastic expectation-based calibration approaches to better exploit the behavioral structure of models like PT.

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DATA AVAILABILITY

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STATEMENT ON THE USE OF AI TOOLS

Artificial intelligence tools were used solely to support the writing and editing process of this paper. No AI tools were used to generate original content or research ideas; all research, analysis, and conclusions are the work of the authors.

AUTHOR CONTRIBUTIONS

The authors confirm contribution to the paper as follows: conceptualization: Pedram Beigi, Mohammad Emad Rashidi, Nachuan Li, Shayan Bafandkar, Hani Mahmassani, Alireza Talebpour, Samer Hamdar; methodology: Pedram Beigi, Mohammad Emad Rashidi, Nachuan Li, Shayan Bafandkar, Dana Monzer, John Hourdos, Hani Mahmassani, Alireza Talebpour, Samer Hamdar; validation: Pedram Beigi, Mohammad Emad Rashidi, Nachuan Li, Shayan Bafandkar, Dana Monzer; project administration: John Hourdos; funding acquisition: Hani Mahmassani, Alireza Talebpour, Samer Hamdar; supervision: Samer Hamdar; draft manuscript preparation: Pedram Beigi, Mohammad Emad Rashidi, Nachuan Li, Shayan Bafandkar, Dana Monzer, Samer Hamdar; manuscript review and editing: Pedram Beigi, Hani Mahmassani, Alireza Talebpour, Samer Hamdar. All authors reviewed the results and approved the final version of the manuscript.

DECLARATION OF CONFLICTING INTERESTS

The authors declared the following potential conflicts of interest with respect to the research, authorship, and/or publication of this article: Samer Hamdar is a member of the Transportation Research Record's Editorial Board. All other authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

REFERENCES

- [1] E. Chung and A.-G. Dumont, *Transport simulation: beyond traditional approaches*. EPFL press, 2009.
- [2] S.-T. Zheng, M. A. Makridis, A. Kouvelas, R. Jiang, and B. Jia, "A multi-objective calibration framework for capturing the behavioral patterns of autonomously-driven vehicles," *Transp. Res. Part C Emerg. Technol.*, vol. 152, p. 104151, 2023.
- [3] Z. Zheng and M. Sarvi, "Modeling, calibrating, and validating car following and lane changing behavior," *Transp. Res. Part C Emerg. Technol.*, vol. 71, pp. 182–183, 2016.
- [4] P. G. Gipps, "A behavioural car-following model for computer simulation," *Transportation Research Part B*, vol. 15, no. 2, pp. 105–111, 1981, doi: 10.1016/0191-2615(81)90037-0.
- [5] H. M. Zhang and T. Kim, "A car-following theory for multiphase vehicular traffic flow," *Transportation Research Part B: Methodological*, vol. 39, no. 5, pp. 385–399, 2005.
- [6] L. Li, X. M. Chen, and L. Zhang, "A global optimization algorithm for trajectory data based car-following model calibration," *Transp. Res. Part C Emerg. Technol.*, vol. 68, pp. 311–332, 2016.
- [7] V. Punzo and F. Simonelli, "Analysis and comparison of microscopic traffic flow models with real traffic microscopic data," *Transp. Res. Rec.*, vol. 1934, no. 1, pp. 53–63, 2005.
- [8] W. Sleiman *et al.*, "Diffusion Process-Based Model for Network Trajectory Propagation: Formulation, Data Mining, and Cross-Comparative Analysis," *IEEE Transactions on Intelligent Transportation Systems*, 2026.
- [9] K. Aghabayk, M. Sarvi, and W. Young, "A state-of-the-art review of car-following models with particular considerations of heavy vehicles," *Transp. Rev.*, vol. 35, no. 1, pp. 82–105, 2015.
- [10] S. H. Hamdar, L. Qin, and A. Talebpour, "Weather and road geometry impact on longitudinal driving behavior: Exploratory analysis using an empirically supported acceleration modeling framework," *Transp. Res. Part C Emerg. Technol.*, vol. 67, pp. 193–213, 2016.
- [11] S. Feng, X. Yan, H. Sun, Y. Feng, and H. X. Liu, "Intelligent driving intelligence test for autonomous vehicles with naturalistic and adversarial environment," *Nat. Commun.*, vol. 12, no. 1, p. 748, 2021.
- [12] A. Alhariqi, Z. Gu, and M. Saberi, "Calibration of the intelligent driver model (idm) with adaptive parameters for mixed autonomy traffic using experimental trajectory data," *Transportmetrica B: transport dynamics*, vol. 10, no. 1, pp. 421–440, 2022.
- [13] R. Ammourah *et al.*, "Introduction to the Third Generation Simulation Dataset: Data Collection and Trajectory Extraction," *Transp. Res. Rec.*, 2024, doi: 10.1177/03611981241257257.
- [14] M. Treiber, A. Hennecke, and D. Helbing, "Congested traffic states in empirical observations and microscopic simulations," *Phys. Rev. E*, vol. 62, no. 2, p. 1805, 2000.

- [15] S. H. Hamdar and H. S. Mahmassani, "From existing accident-free car-following models to colliding vehicles: exploration and assessment," *Transp Res Rec*, vol. 2088, no. 1, pp. 45–56, 2008.
- [16] M. Zhu, X. Wang, A. Tarko, and others, "Modeling car-following behavior on urban expressways in Shanghai: A naturalistic driving study," *Transp Res Part C Emerg Technol*, vol. 93, pp. 425–445, 2018.
- [17] M. Pourabdollah, E. Bjarkvik, F. Furer, B. Lindenberg, and K. Burgdorf, "Calibration and evaluation of car following models using real-world driving data," *IEEE Conference on Intelligent Transportation Systems, Proceedings, ITSC*, vol. 2018-March, no. June 2018, pp. 1–6, 2018, doi: 10.1109/ITSC.2017.8317836.
- [18] C. Chen, L. Li, J. Hu, and C. Geng, "Calibration of MITSIM and IDM car-following model based on NGSIM trajectory datasets," *Proceedings of 2010 IEEE International Conference on Vehicular Electronics and Safety, ICVES 2010*, pp. 48–53, 2010, doi: 10.1109/ICVES.2010.5550943.
- [19] A. Adewale and C. Lee, "Prediction of car-following behavior of autonomous vehicle and human-driven vehicle based on drivers' memory and cooperation with lead vehicle," *Transp. Res. Rec.*, vol. 2678, no. 6, pp. 248–266, 2024.
- [20] S. Ding, X. Chen, Z. Fu, and F. Peng, "An Extended Car-Following Model in Connected and Autonomous Vehicle Environment: Perspective From the Cooperation between Drivers," *J. Adv. Transp.*, vol. 2021, p. 2739129, 2021.
- [21] A. Sharma, Z. Zheng, A. Bhaskar, and M. M. Haque, "Modelling Car-Following Behaviour of Connected Vehicles With a Focus on Driver Compliance," *Transportation Research Part B: Methodological*, vol. 126, pp. 256–279, 2019.
- [22] S. H. Hamdar, H. S. Mahmassani, and M. Treiber, "From behavioral psychology to acceleration modeling: Calibration, validation, and exploration of drivers' cognitive and safety parameters in a risk-taking environment," *Transportation Research Part B: Methodological*, vol. 78, pp. 32–53, 2015.
- [23] T. Li, D. Chen, H. Zhou, J. Laval, and Y. Xie, "Car-following behavior characteristics of adaptive cruise control vehicles based on empirical experiments," *Transportation Research Part B: Methodological*, vol. 147, pp. 67–91, 2021, doi: 10.1016/j.trb.2021.03.003.
- [24] G. F. Newell, "A simplified car-following theory: A lower order model," *Transportation Research Part B: Methodological*, vol. 36, no. 3, pp. 195–205, 2002, doi: 10.1016/S0191-2615(00)00044-8.
- [25] S. Lapardhaja, K. U. Yagantekin, M. Yang, T. A. Majumder, X. Kan, and M. Badhrudeen, "Unlocking potential capacity benefits of electric vehicles (EVs) with adaptive cruise control (ACC)," *Transportmetrica B*, vol. 11, no. 1, pp. 1894–1911, 2023, doi: 10.1080/21680566.2023.2271668.
- [26] A. Zare, M. Shang, X. D. Kan, and R. Stern, "The Electric Vehicle Model (EVM): A Novel Car-Following Model for Electric Adaptive Cruise Control Vehicles," *2024 Forum for Innovative Sustainable Transportation Systems, FISTS 2024*, pp. 1–6, 2024, doi: 10.1109/FISTS60717.2024.10485590.
- [27] B. Kim and K. P. Heaslip, "Identifying suitable car-following models to simulate automated vehicles on highways," *International Journal of Transportation Science and Technology*, vol. 12, no. 2, pp. 652–664, 2023, doi: 10.1016/j.ijtst.2023.02.003.

- [28] J. Bai, S. Mao, and J. J. Lee, "A novel car-following model for adaptive cruise control vehicles using enhanced intelligent driver model," *Transportation Letters*, vol. 00, no. 00, pp. 1–17, 2024, doi: 10.1080/19427867.2024.2376409.
- [29] M. Makridis, K. Mattas, A. Anesiadou, and B. Ciuffo, "OpenACC. An Open Database of Car-Following Experiments to Study the Properties of Commercial ACC Systems," *Transp. Res. Part C Emerg. Technol.*, vol. 125, p. 103047, Apr. 2021, doi: 10.1016/j.trc.2021.103047.
- [30] C. Li, D. Huang, T. Wang, and J. Qin, "Lane-changing decision rule with the difference of traffic flow's variation in multi-lane highway for connected and autonomous vehicles," *Transportation safety and environment*, vol. 5, no. 3, 2023.
- [31] X. Hu, Z. Zheng, D. Chen, and J. Sun, "Autonomous Vehicle's Impact on Traffic: Empirical Evidence From Waymo Open Dataset and Implications From Modelling," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 6, pp. 6711–6724, 2023, doi: 10.1109/TITS.2023.3258145.
- [32] P. Beigi, S. Hamdar, A. Talebpour, and H. Mahmassani, "Impact of Heavy Vehicles on Headway Distributions: Study Using Naturalistic Urban Expressway Trajectories," *Transp. Res. Rec.*, vol. 2679, no. 7, pp. 222–235, 2025.
- [33] M. Treiber and A. Kesting, "Modeling lane-changing decisions with mobil," in *Traffic and Granular Flow'07*, Springer, 2009, pp. 211–221.
- [34] A. Kesting, M. Treiber, and D. Helbing, "Enhanced intelligent driver model to access the impact of driving strategies on traffic capacity," *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, vol. 368, no. 1928, pp. 4585–4605, 2010, doi: 10.1098/rsta.2010.0084.
- [35] S. H. Hamdar, M. Treiber, H. S. Mahmassani, and A. Kesting, "Modeling Driver Behavior as a Sequential Risk Taking Task," *Transp. Res. Rec.*, vol. 2088, pp. 208–217, 2008.
- [36] S. H. Hamdar, H. S. Mahmassani, and M. Treiber, "From behavioral psychology to acceleration modeling: Calibration, validation, and exploration of drivers' cognitive and safety parameters in a risk-taking environment," *Transportation Research Part B: Methodological*, vol. 78, pp. 32–53, 2015.
- [37] E. Brockfeld, R. D. Kühne, and P. Wagner, "Calibration and validation of microscopic traffic flow models," *Transp. Res. Rec.*, vol. 1876, no. 1, pp. 62–70, 2004.
- [38] A. Kesting, M. Treiber, and D. Helbing, "Enhanced intelligent driver model to access the impact of driving strategies on traffic capacity," *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, vol. 368, no. 1928, pp. 4585–4605, 2010.
- [39] B. L. Welch, "On the comparison of several mean values: an alternative approach," *Biometrika*, vol. 38, no. 3/4, pp. 330–336, 1951.
- [40] P. A. Games and J. F. Howell, "Pairwise multiple comparison procedures with unequal n's and/or variances: a Monte Carlo study," *Journal of Educational Statistics*, vol. 1, no. 2, pp. 113–125, 1976.